

Assignment 3

Find the Mean of the following:

1) 10, 20, 30, 40, 50

$$= 10 + 20 + 30 + 40 + 50 / 5$$

$$= 150 / 50$$

$$= 30$$

2) 5, 10, 15, 20

$$= 5 + 10 + 15 + 20 / 4$$

$$= 50 / 4$$

$$= 12.5$$

3) 12, 18, 24, 30, 36

$$= 12 + 18 + 24 + 30 + 36 / 5$$

$$= 120 / 5$$

$$= 24$$

4) 8, 16, 24, 32

$$= 8 + 16 + 24 + 32 / 4$$

$$= 80 / 4$$

$$= 20$$

5) 25, 35, 45, 55, 65

$$= 25 + 35 + 45 + 55 + 65 / 5$$

$$= 225 / 5$$

$$= 45$$

Find the Median of following:

1) 10, 20, 30, 40, 50

$$= 30$$

2) 5, 10, 15, 20

$$= 10 + 15 / 2$$

$$= 25 / 2$$

$$= 12.5$$

$$3) 12, 18, 24, 30, 36$$

$$= 24$$

$$4) 7, 14, 21, 28$$

$$= 14+21 / 2$$

$$= 35 / 2$$

$$= 17.5$$

$$5) 11, 22, 33, 44, 55$$

$$= 33$$

Find the Mode of the following:

$$1) 10, 20, 20, 30, 40$$

$$= 20$$

$$2) 5, 5, 10, 15, 20$$

$$= 5$$

$$3) 7, 8, 8, 9, 10$$

$$= 8$$

$$4) 25, 25, 30, 35, 40$$

$$= 25$$

$$5) 1, 2, 2, 3, 4$$

$$= 2$$

Find the Variance of the following:

$$1) [2, 4, 6]$$

$$= 2.67$$

$$2) [10, 20, 30]$$

$$= 66.67$$

$$3) [5, 5, 5]$$

$$= 0$$

$$4) [1, 3, 5]$$

$$= 2.67$$

$$5) [4, 8, 12]$$

$$= 10.67$$

Find the standard deviation of the following:

$$1. [2, 4, 6]$$

$$= 1.63$$

$$2. [10, 20, 30]$$

$$= 8.16$$

$$3. [5, 5, 5]$$

$$= 0$$

$$4. [1, 3, 5]$$

$$= 1.63$$

$$5. [4, 8, 12]$$

$$= 3.27$$

Find Q1, Q2, and Q3 for:

$$1) [10, 20, 30, 40, 50, 60, 70, 80]$$

$$= Q1:25, Q2:45, Q3:65$$

$$2) [5, 10, 15, 20, 25, 30, 35, 40]$$

$$= Q1:12.5, Q2:22.5, Q3:32.5$$

3) [2, 4, 6, 8, 10, 12, 14, 16]

= Q1:5, Q2:9, Q3:13

4) [1, 2, 3, 4, 5, 6, 7, 8]

= Q1:2.5, Q2:4.5, Q3:6.5

5) [12, 24, 36, 48, 60, 72, 84, 96]

= Q1:30, Q2:54, Q3:78

Find the percentiles for the following:

1) 25th Percentile of [10, 20, 30, 40, 50]

= 20

2) 50th Percentile of [5, 10, 15, 20, 25]

= 15

3) 75th Percentile of [2, 4, 6, 8, 10]

= 8

4) 90th Percentile of [10, 20, 30, 40, 50]

= 50

5) 60th Percentile of [12, 24, 36, 48, 60]

= 36